Université d'Ottawa · University of Ottawa
Faculté des sciences
Mathématiques et de statistique
Faculty of Science
Mathematics and Statistics

MAT 1302 B: Mathematical methods II

Instructor: Aziz Khanchi

Test II - Version 1 - Solutions

February 2011

Surname _____ First Name _____

Student # _____ DGD (1–4) _____

Instructions:

- No calculators or other electronic aids allowed.
- No notes, books or other papers allowed.
- Answer all questions in the space provided. You must justify your answers and explain your reasoning.
- In all there are 8 questions worth a total of 30 marks. There are 4 multiple choice questions, and 4 long answer questions. There will be no part marking for the multiple choice questions. For the long answer questions, you must justify your answers.

GOOD LUCK!

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Do not write anything in this table.

Q1-Q4	Q5	Q6	Q7	Q8	Total
/8	/7	/5	/5	/5	/30

Part 1: Multiple choice questions, 2 points each (no partial marks), choose the best possible answer.

Question 1. For what values of h the following vectors are independent?

$$\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ h \\ 0 \end{bmatrix}$$

A) For all values of h,

B) $h = 1$,

C) $h \neq 1$,

D) $h \neq 0$,

E) $h = 0$.

Solution: The three vectors are independent if equation

$$x_1 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} + x_3 \begin{bmatrix} 0 \\ h \\ 0 \end{bmatrix} = 0$$

has only the trivial solution.

$$\begin{bmatrix} cx_1 \\ x_1 + x_2 + hx_3 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \Longrightarrow x_1 = x_2 = 0,$$

For this equation to have only the trivial solution h must be non-zero.

Question 2. Consider $A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 3 \\ 1 & 0 \end{bmatrix}$. Determine $AI_2(2B^T) + B$.

A) $\begin{bmatrix} 18 & 5 \\ 7 & 0 \end{bmatrix}$,

B) $\begin{bmatrix} 10 & 4 \\ 3 & 0 \end{bmatrix}$,

C) $\begin{bmatrix} 1 & 3 \\ 3 & 4 \end{bmatrix}$,

D) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$,

E) at least one of the operations is not well defined.

Solution: A

Question 3. Suppose A is an $n \times n$ invertible matrix. Which statement is **false**?

A) A^{-1} is invertible,

B) $A\overrightarrow{x} = \overrightarrow{0}$ is consistent,

C) If B is invertible then AB is always invertible,

D) There exists a vector $\overrightarrow{b}$ in $\mathbb{R}^n$ such that $A\overrightarrow{x} = \overrightarrow{b}$ is inconsistent.

E) A^T is invertible.

Solution: D

Question 4. Suppose $A = PB$, where $P = \begin{bmatrix} 1 & 1 \\ 2 & 3 \end{bmatrix}$ and $A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$. The first row of B is:

A) $[\ 5 \quad 8\ ]$,

B) $[\ 5 \quad -3\ ]$,

C) $[\ -1 \quad 3\ ]$,

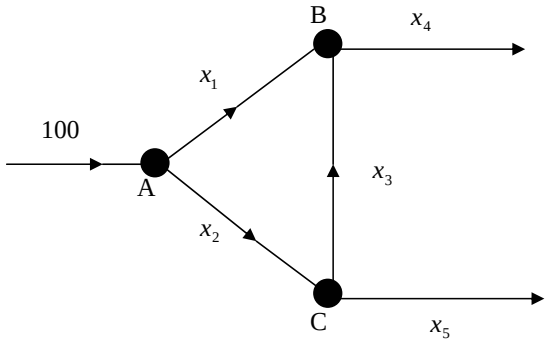
D) $[\ 1 \quad -2\ ]$,

E) None of the preceding.

Solution: Note that $B = P^{-1}A$, $\det P = (3)(1) - (2)(1) = 1$ and $P^{-1} = \frac{1}{1}\begin{bmatrix} 3 & -1 \\ -2 & 1 \end{bmatrix}$. Therefore, $B = P^{-1}A = P^{-1} = \frac{1}{1}\begin{bmatrix} 3 & -1 \\ -2 & 1 \end{bmatrix}\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} -1 & 3 \\ 1 & -2 \end{bmatrix}$. C is correct.

Part 2: Long Answer Questions (Details are necessary!)

Question 5. (7 points) Consider the following traffic network with one sided streets.



A) Write the linear system associated with the network flow. (Do not solve the system.)

Solution:

	input	=	output
Total network:	100	=	$x_4 + x_5$
Node A:	100	=	$x_1 + x_2$
Node B:	$x_1 + x_3$	=	x_4
Node C:	x_2	=	$x_3 + x_5$

$$\begin{cases} x_4 + x_5 = 100 \\ x_1 + x_2 = 100 \\ x_1 + x_3 - x_4 = 0 \\ x_2 - x_3 - x_5 = 0 \end{cases}$$

B) Suppose that the augmented matrix of the linear system in part A is reduced to

$$\left[\begin{array}{ccccc|c} 1 & 0 & 1 & 0 & 1 & 100 \\ 0 & 1 & -1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 100 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right].$$

Give the general pattern of the traffic flow, i.e. the general solution of the linear system. At this stage ignore the constraints.

Solution: x_3 and x_5 are free variables. The general solution is $\begin{cases} x_1 = 100 - x_3 - x_5 \\ x_2 = x_3 + x_5 \\ x_4 = 100 - x_5 \\ x_3, x_5 \text{ free} \end{cases}$

C) If we close the street associated with x_5, find the maximum of x_1.

Solution: If $x_5 = 0$, the general solution changes to $\begin{cases} x_1 = 100 - x_3 \\ x_2 = x_3 \\ x_4 = 100 \\ x_5 = 0 \\ x_3, \text{ free} \end{cases}$

The constraint $x_3 \geq 0$ implies that $x_1 = 100 - x_3 \leq 100$. The maximum of x_1 is 100.

Question 6. (5 points) A closed economy has two sectors telecommunication and transportation. Telecommunication sells 80% of its output to transportation and retains the rest. Moreover, transportation sells 30% of its output to telecommunication and retains the rest.

A) Construct the input-output table.

Solution:

tel.	trans.	
0.2	0.3	tel.
0.8	0.7	trans.

B) Write down the linear system corresponding to this economy, which gives the equilibrium prices.

Solution: p_1: price of telecommunication, p_2: price of transportation.

$$\begin{cases} 0.2p_1 + 0.3p_2 = p_1 \\ 0.8p_1 + 0.7p_2 = p_2. \end{cases} \Longrightarrow \begin{cases} -0.8p_1 + 0.3p_2 = 0 \\ 0.8p_1 - 0.3p_2 = 0. \end{cases}$$

C) Find the equilibrium price of telecommunication when the price of transportation is 100 units.

Solution: From both equations $p_1 = \frac{3}{8}p_2$. If $p_2 = 100$, then $p_1 = \frac{300}{8}$.

Question 7. (5 points) Consider the following matrix

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 5 & 2 \\ 3 & 11 & 2 \end{bmatrix}.$$

A) Find the inverse of the matrix A. (Hint: The inverse has only integer entries.)

Solution:

$$\left[A \mid I\right] = \left[\begin{array}{ccc|ccc} 1 & 2 & 1 & 1 & 0 & 0 \\ 2 & 5 & 2 & 0 & 1 & 0 \\ 3 & 11 & 2 & 0 & 0 & 1 \end{array}\right] \xrightarrow[-3R_3+R_3\to R_3]{-2R_1+R_2\to R_2} \left[\begin{array}{ccc|ccc} 1 & 2 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & -2 & 1 & 0 \\ 0 & 5 & -1 & -3 & 0 & 1 \end{array}\right] \xrightarrow[-5R_2+R_3\to R_3]{-2R_2+R_1\to R_1}$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 5 & -2 & 0 \\ 0 & 1 & 0 & -2 & 1 & 0 \\ 0 & 0 & -1 & 7 & -5 & 1 \end{array}\right] \xrightarrow[-R_3\to R_3]{R_3+R_1\to R_1} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 12 & -7 & 1 \\ 0 & 1 & 0 & -2 & 1 & 0 \\ 0 & 0 & 1 & -7 & 5 & -1 \end{array}\right]$$

Hence, A is invertible and $A^{-1} = \begin{bmatrix} 12 & -7 & 1 \\ -2 & 1 & 0 \\ -7 & 5 & -1 \end{bmatrix}$

B) Use A^{-1} to solve $AX = \begin{bmatrix} 3 \\ 0 \\ 1 \end{bmatrix}$.

Solution: $X = A^{-1} \begin{bmatrix} 3 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 12 & -7 & 1 \\ -2 & 1 & 0 \\ -7 & 5 & -1 \end{bmatrix} \begin{bmatrix} 3 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 37 \\ -6 \\ -22 \end{bmatrix}$.

Question 8. (5 points) Find two vectors spanning the solution set of the following equation:

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 0 & 0 & 0 \end{bmatrix} \overrightarrow{x} = \overrightarrow{0}.$$

Solution: The corresponding augmented matrix is

$$\begin{bmatrix} 1 & 2 & 3 & |0 \\ 2 & 4 & 6 & |0 \\ 0 & 0 & 0 & |0 \end{bmatrix} \xrightarrow{-2R_1+R_2 \to R_2} \begin{bmatrix} 1 & 2 & 3 & |0 \\ 0 & 0 & 0 & |0 \\ 0 & 0 & 0 & |0 \end{bmatrix}$$

The general solution is $\begin{cases} x_1 = -2x_2 - 3x_3 \\ x_2, x_3 \quad \text{are free} \end{cases}$. Write the solution in parametric vector form:

$$\begin{bmatrix} -2x_2 - 3x_3 \\ x_2 \\ x_3 \end{bmatrix} = x_2 \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix}.$$

Therefore, solution set $= \text{span} \left\{ \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix} \right\}$.

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